○ Decision Making
● Information Sharing
○ Directions

MJCF XML, dtDynSpec, dtDynModel, Runtime Update

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Joonhee Jo

dtDynamics 0.1.3 checked setup

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오늘의 흐름

01 Architecture — XML에서 runtime까지 세 layer

02 XML subset — typed code, exact path, zero-based offset

03 Compilation — ValidateSpec와 CompileSpecChecked

04 LeoQuad runtime — state staging, dynamics, export

05 Runtime update — 안전한 reload와 hot-swap

Part 1 — Architecture and layers

The whole path

MJCF-like XML

Robot description file. Import/export format only.

→

dtDynSpec

`DecodeXml` builds an editable model tree.

→

dtDynModel

`CompileSpecChecked` validates external/editable input, then freezes indices, dimensions, and layout.

→

dtDynData

`Resize(model)` allocates runtime buffers once.

every control tick

qpos/qvel update

`ComputeDynamicsCCRBA` + `ComputeJacobians`

→

WBC fields

Controller-facing poses, Jacobians, centroidal terms.

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Why MJCF-like XML?

`dtDynamics` is **not** MuJoCo and is not a MuJoCo wrapper. It reuses useful model semantics:

- body tree rooted at `world`
- free, ball, slide, and hinge joints
- MuJoCo-compatible `qpos` and `qvel` sizes
- named body, joint, site, and actuator lookup
- MJCF-style robot description files
- external validation against MuJoCo and Pinocchio

Boundary

Production code remains `dtMath`-only. The XML dialect is a file-format choice, not a runtime dependency.

LeoQuad package boundary

LeoQuad links the installed native package:

```
find_package(dtDynamics 0.1.3 EXACT CONFIG REQUIRED)
target_link_libraries(dtcontrol PUBLIC dt::Dynamics)
```

Important consequence

- `<mujoco>` can still be the XML root.
- It is only a file-format choice.
- The native package does not include MuJoCo headers.
- The native package does not link MuJoCo libraries.

Three runtime layers

Layer	Mutable?	Purpose	Use it when…
`dtDynSpec`	yes	editable model description	importing, editing, exporting XML
`dtDynModel`	fixed topology; `UpdateSpec` only	compiled index/layout contract	control code needs stable IDs and dimensions
`dtDynData`	yes	runtime state, scratch, outputs	every control tick updates state and recomputes

Key design rule — Topology/layout/name-map changes require checked recompilation and data resize. Fixed-layout physical parameter changes use `UpdateSpec` only after validation and with exclusive model access at a reader-free control-safe boundary. An incompatible spec returns false without model writes; after success, recompute stale outputs before consumption. State changes only require `qpos/qvel` update plus recompute. External/editable `dtDynSpec` exposes an immutable collection view; use `EditBody` / `EditJoint` / `EditSite` / `EditActuator` only during setup.

dtDynSpec: editable tree

Owns dynamic containers for

- bodies
- joints
- sites
- actuators
- inertial properties
- parent-child topology

Operational details

- `Clear()` creates body `0` named `world`.
- `AddBody(name, parent)` appends a body.
- A body with no joints is fixed.
- Joints and sites attach to their owning body.
- `DecodeXml(xml)` imports the supported MJCF-like subset.
- `EncodeXml()` writes deterministic canonical XML.

dtDynModel: compiled contract

`dtDynModel` answers the questions control code cares about:

- how many generalized coordinates exist: `nq`
- how many velocities exist: `nv`
- where each joint starts in `qpos`
- where each joint starts in `qvel`
- which body owns each DOF
- which bodies are fixed
- which bodies are roots
- where names map to body, joint, site, and actuator IDs
- how large all `dtDynData` buffers must be

> After compilation, controller code should treat it as **read-only**.

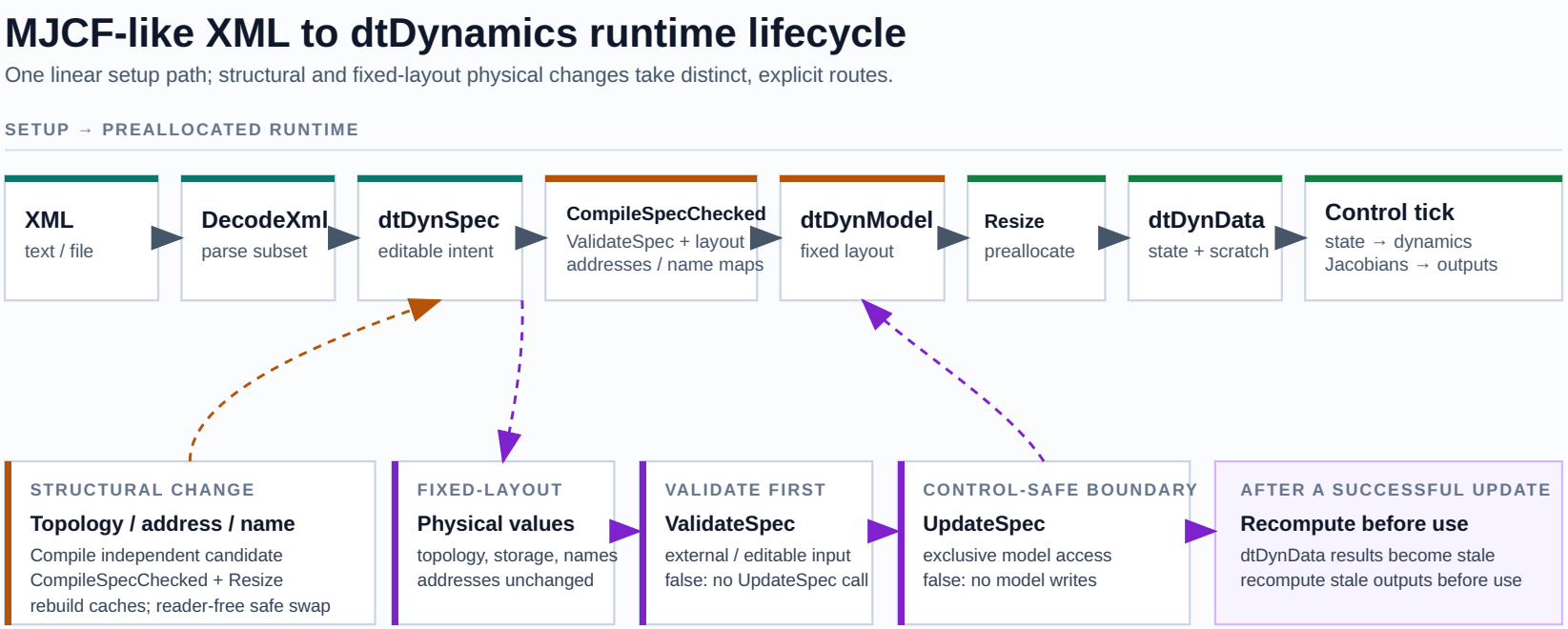
dtDynData: runtime state and output

`dtDynData::Resize(model)` allocates buffers for one model shape. It stores:

- input state: `qpos`, `qvel`
- body pose outputs: `xpos`, `xmat`, `xipos`
- site pose outputs: `site_xpos`, `site_xmat`
- Jacobians: body and site linear/angular terms plus dot terms
- dynamics: `Mmat`, `Cmat`, gravity, coriolis, nonlinear torque
- centroidal terms: `cmAmat`, `centMom`, `comPos`, `comVel`
- scratch arrays used by recursive dynamics algorithms

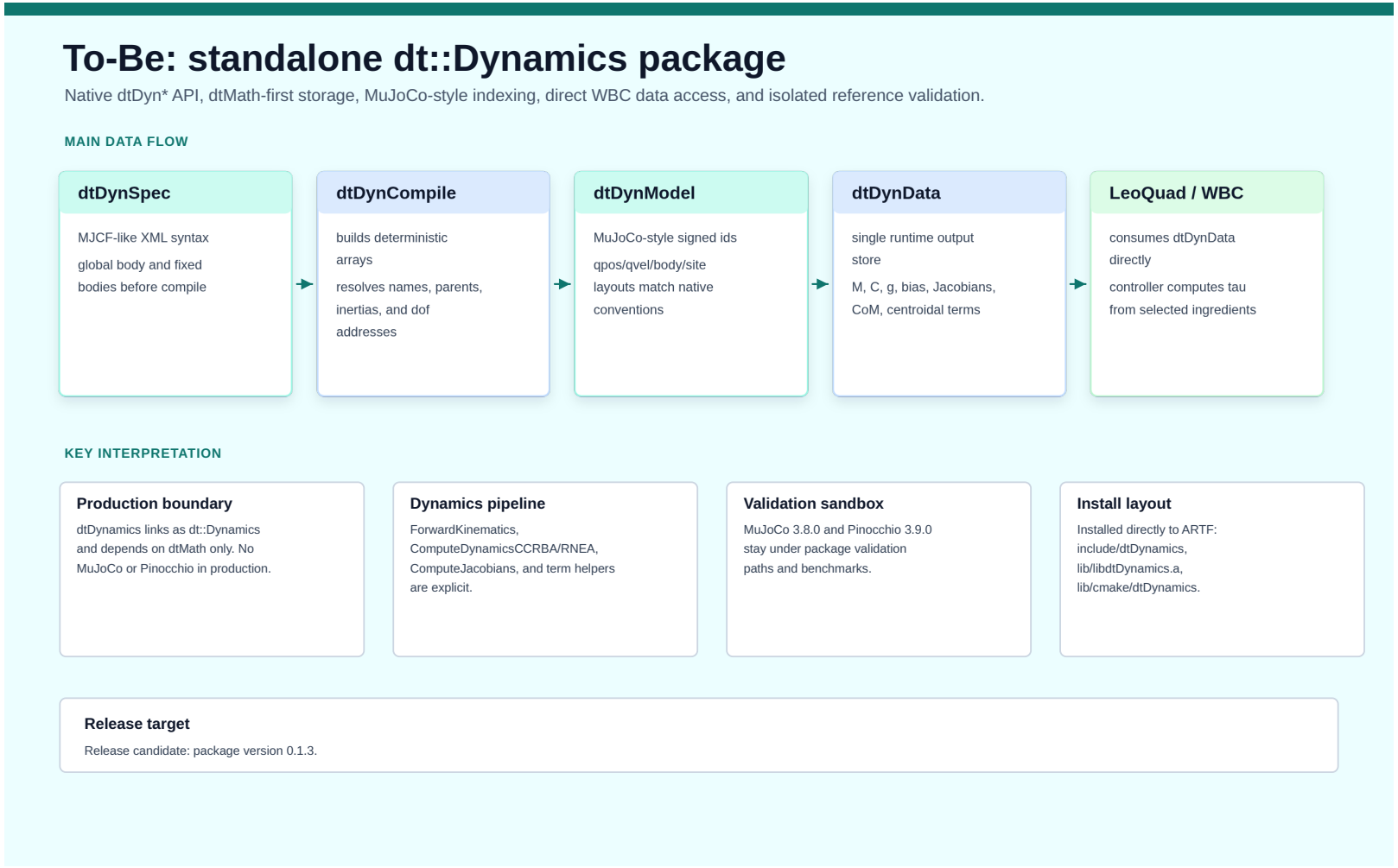
> Real-time ticks update **values**, not **shapes**.

Object lifetime diagram



To-Be standalone dtDynamics architecture

Higher-level package view; not a literal diagram of the current LeoQuad export-copy path.



Part 2 — XML subset and decode

Supported XML subset

XML item	Native behavior
`<mujoco>`	root syntax only
`<compiler angle="radian">`	no scaling
`<compiler angle="degree">`	range conversion to radians
other compiler angle units	throw an unsupported-angle error
XML comments	valid comments are skipped; malformed comments throw
`<worldbody>`	enables body-tree parsing
`<body name pos quat>`	creates `dtDynBodySpec`
`<inertial pos quat mass diaginertia>`	fills inertial spec
`<joint name type pos axis range limited armature>`	creates joint spec; omitted `type` follows the documented hinge default, unknown values fail
`<freejoint name>`	creates free joint
`<site name pos quat>`	creates task/sensor frame metadata
`<motor name joint gear>`	maps actuator to parsed joint

Unsupported XML fails closed

Explicitly rejected

XML item	Behavior
`<equality>`, `<tendon>`	unsupported tree-only dynamics error
`<asset>`, `<mesh>`, `<geom>`	unsupported element error; no silent rendering-only skip
unknown elements	typed unsupported diagnostic at the original byte offset

Accepted metadata boundary

- documented `<option>` metadata used by validation fixtures is preserved
- only documented attributes have native model semantics; extra metadata is not a dynamics contract
- unsupported simulator rendering fields are rejected rather than silently discarded
- invalid numbers, wrong vector arity, duplicate names, and unresolved references fail

`dtDynamics 0.1.3`의 XML은 arbitrary MJCF importer가 아니라 documented tree-model subset이다.

DecodeXml mental model

`dtDynSpec::DecodeXml(xml)` parses XML already in memory. `dtDynSpec::LoadXmlFile(path)` reads a file and delegates to the same decoder with path-aware diagnostics. 실패 시 `dtDynXmlError::Code()`, exact `SourcePath()`, original input의 zero-based `Offset()`을 machine interface로 사용한다.

1. Reject `<equality>` and `<tendon>`.
2. Create a fresh spec with `world` at body index `0`.
3. Track whether parsing is inside `<worldbody>` or `<actuator>`.
4. Track a body stack for nested body parentage.
5. Read `<compiler angle="...">` for range scaling.
6. Parse supported tags into spec containers.
7. Resolve motor actuators by joint name.
8. Return the editable `dtDynSpec`.

Minimal decode example

```xml
<mujoco model="single_arm">
  <compiler angle="radian"/>
  <worldbody>
    <body name="base" pos="0 0 0">
      <body name="link" pos="1 0 0">
        <inertial pos="0.5 0 0" mass="2.5"
                  diaginertia="0.1 0.2 0.3"/>
        <joint name="hinge_z" type="hinge"
               axis="0 0 1" armature="0.05"/>
        <site name="tool" pos="1 0 0"/>
      </body>
    </body>
  </worldbody>
  <actuator>
    <motor name="hinge_motor" joint="hinge_z" gear="80"/>
  </actuator>
</mujoco>
```

Decoded structure

```
world
 -> base   (fixed body)
 -> link   (hinge body)
 -> tool   (site)
hinge_motor -> hinge_z, gear=80
```

LeoQuad XML subset

In `model/LeoQuad_QD1.xml`:

```
<body name="BODY">
  <freejoint name="BODY" />
  <inertial pos="0.0056 0 0.0064"
            quat="0 0.707107 0 0.707107"
            mass="18.0916"
            diaginertia="0.5007 0.4487 0.0965"/>
  <body name="FL_HP" pos="...">
    <joint name="FL_HR_JOINT" type="hinge" .../>
    ...
    <site name="FL_FOOT" pos="0 0 -0.3244"/>
  </body>
</body>
```

For dtDynamics

- `BODY` is the physical root body under `world`.
- `<freejoint>` gives `BODY` 7 `qpos` and 6 `qvel` entries.
- hip, thigh, and calf joints add one `qpos` and one `qvel` each.
- foot sites become named task frames owned by their nested leg bodies.
- unsupported mesh and geom tags are rejected; collision/render assets remain outside this XML subset.

EncodeXml role

`dtDynSpec::EncodeXml()` writes deterministic XML for the supported subset.

It emits

- `<mujoco>` root
- `<compiler angle="radian"/>`
- recursive `<worldbody>` body tree
- body `name`, `pos`, and `quat`
- inertial data when `mass != 0`
- all joints as `<joint ... type="...">`; decoded `<freejoint>` becomes `type="free"`
- sites with pose attributes
- actuator motors if actuator specs exist
- floating-point values at high precision

Native model-spec export

Use it as a **native model-spec exporter**, not as a full MuJoCo scene exporter.

```
spec.SaveXmlFile("model/robot-native.xml");
```

`SaveXmlFile` writes the exact `EncodeXml()` bytes; create parent directories before calling it.

Round-trip check: `DecodeXml → EncodeXml → DecodeXml` should preserve body, joint, site, and actuator counts.

Part 3 — Compilation and layout

CompileSpec purpose

`CompileSpecChecked(spec)`는 editable model을 검증한 뒤 fixed array와 address를 만든다. 이미 별도 검증이 끝난 trusted setup data만 raw `CompileSpec(spec, model)`을 사용한다.

Compilation assigns:

- dimensions: `nq`, `nv`, `nbody`, `njnt`, `nsite`, `nu`, `nroot`
- body arrays: parent, joint/DOF address, `root_bodyid`, `root_mass`, `body_rootid`, `body_rootidx`
- joint arrays: type, `qpos` address, `qvel` address
- DOF arrays: body owner, parent DOF, joint owner, armature
- `site_bodyid`, `actuator_jointid`, and `actuator_gear`
- lookup maps for bodies, joints, and sites

This is the transition from **descriptive structure** to **runtime contract**.

Checked setup boundary

외부 XML이나 callback-scoped `Edit*`를 거친 editable spec에는 `ValidateSpec(spec, &error)` 또는 `CompileSpecChecked(spec)`를 사용한다. `dtDynValidationError`의 `code`, `object`, `index`는 machine-readable setup diagnostic이다. validation과 compilation은 control loop 밖에서 수행한다.

```
dtDynModel model = CompileSpecChecked(spec);
dtDynData data;
data.Resize(model);
assert(data.IsCompatible(model));
```

`IsCompatible()`은 storage dimension만 확인하며 model identity를 증명하지 않는다.

Joint sizes

`dtDynamics` follows MuJoCo-compatible sizes.

Joint type	Enum	`qpos`	`qvel`
Free	0	7	6
Ball	1	4	3
Slide	2	1	1
Hinge	3	1	1

LeoQuad layout

```
qpos = [base position_G(3),
        base quaternion_GB(4),
        12 joint positions]
     = 19 values

qvel = [base linear velocity_G(3),
        base angular velocity_B(3),
        12 joint velocities]
     = 18 values
```

LeoQuad dimension guard

LeoQuad checks the compiled model before using it:

```
if (model.dynModel.nq != DOF_TOTAL_QUAT || model.dynModel.nv != DOF_TOTAL) {
    throw std::runtime_error(
        "Robot::InitializeDynamicsModel model dimensions do not match LeoQuad qpos/qvel constants");
}
```

Why this matters

- XML structure controls `nq` and `nv`.
- Controller constants assume one floating base plus 12 actuated joints.
- Any structural model update must preserve that contract or intentionally update the controller side too.

qpos/qvel addresses

For a free base plus one hinge:

```
joint 0: floating_base
  qpos adr = 0    qpos size = 7
  qvel adr = 0    qvel size = 6

joint 1: hip_pitch
  qpos adr = 7    qpos size = 1
  qvel adr = 6    qvel size = 1
```

First hinge after a free base

```
data.qpos(7) = hipPosition;
data.qvel(6) = hipVelocity;
```


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World, fixed bodies, and roots

World body

- body ID `0`
- not physical mass
- no joint and no DOF
- topology parent for physical roots
- not a dynamics-recursion participant

Fixed body

- any non-world body with no joint
- can have mass, inertia, sites, and child bodies
- contributes to composite inertia
- does not create `qpos` or `qvel` columns

Root metadata identifies each physical root and its mass. For multi-root models, `cmAmat` is `6*nroot × nv` and `comJmat` is `3*nroot × nv`.

Part 4 — LeoQuad runtime loop

LeoQuad initialization path

`Robot::InitializeDynamicsModel(model, data, fileName)` does:

```
const dtDynSpec spec =
    dtDynSpec::LoadXmlFile(fileName);
model.dynModel = dt::Dynamics::CompileSpecChecked(spec);
data.dynData.Resize(model.dynModel);
```

Then it validates and caches IDs

- required body names resolve through `BodyId(...)`
- required task names resolve through `SiteId(...)`
- task body IDs are derived from `site_bodyid`
- Jacobian body IDs include the root/base body
- Jacobian site IDs include all task sites

Cached IDs prevent repeated name lookup in the control loop.

Per-cycle state staging

Every `Robot::UpdateDynamics` writes the current state:

```
// qpos: base position in world.
data.dynData.qpos(0) = data.actControlState.basePosG_G(0);
data.dynData.qpos(1) = data.actControlState.basePosG_G(1);
data.dynData.qpos(2) = data.actControlState.basePosG_G(2);

// qpos: base quaternion.
for (SYSINDEX i = 0; i < DOF4; ++i) {
    data.dynData.qpos(DOF3 + i) = data.base.quat(i);
}

// qpos/qvel: joints after floating base.
for (SYSINDEX i = 0; i < NUM_JOINT; ++i) {
    data.dynData.qpos(DOF7 + i) = data.actJointData.pos_rad(i);
    data.dynData.qvel(DOF6 + i) = data.actJointData.vel_rps(i);
}
```

Velocity convention

LeoQuad uses one fixed convention:

```
qvel = [base linear velocity_G,
        base angular velocity_B,
        joint velocity]
```

There is **no public coordinate-type selector**.

The base fields are staged as:

```
auto &qvel = data.dynData.qvel;
const auto &state = data.actControlState;

// global-frame base linear velocity
qvel(0) = state.baseLinVelB_G(0);
qvel(1) = state.baseLinVelB_G(1);
qvel(2) = state.baseLinVelB_G(2);

// body-frame base angular velocity
for (SYSINDEX i = 0; i < DOF3; ++i) {
    qvel(DOF3 + i) = data.base.angVel_B(i);
}
```

Dynamics and Jacobians per tick

LeoQuad computes a reduced WBC term set:

```
constexpr dt::Dynamics::dtDynCcrbaTermFlags dynamicsTerms =
    dt::Dynamics::DTDYN_CCRBA_TERM_COM_POS |
    dt::Dynamics::DTDYN_CCRBA_TERM_CMAMAT |
    dt::Dynamics::DTDYN_CCRBA_TERM_CENT_MOM;

dt::Dynamics::ComputeDynamicsCCRBA(model.dynModel, data.dynData, gravity, dynamicsTerms);

dt::Dynamics::ComputeJacobians(model.dynModel,
                        data.dynData,
                        model.dynJacobianBodyIds,
                        model.dynJacobianSiteIds);
```

> Separation is intentional: dynamics terms and selected body/site Jacobians are computed through **separate APIs**. Indexed lists avoid computing every body and site Jacobian.

Controller-facing exports

Native field	Controller-facing use
`xpos[bodyId]`	body position
`xipos[bodyId]`	body CoM position
`xmat[bodyId]`	body orientation
`site_xpos[siteId]`	foot/task position
`site_xmat[siteId]`	foot/task orientation
`site_jacp[siteId]`	linear task Jacobian
`site_jacr[siteId]`	angular task Jacobian
`site_jacp_dot[siteId]`	linear task Jacobian derivative
`site_jacr_dot[siteId]`	angular task Jacobian derivative
`body_jacp[baseBodyId]`	base linear Jacobian
`body_jacr[baseBodyId]`	base angular Jacobian
`centMom`, `cmAmat`, `comPos`	centroidal and CoM fields

Part 5 — Runtime model updates

Runtime update has three meanings

State-only update

- same model and buffers
- update `qpos` / `qvel`
- recompute requested dynamics and Jacobians

Fixed-layout physical update

- edit physical values only
- validate off-loop, then `UpdateSpec` with exclusive model access
- `false`: no model writes; recompute stale outputs after success

Structural update

- topology, addresses, names, mapping, or joint type changes
- `CompileSpecChecked`, `Resize`, and rebuild caches
- safe-swap an independent candidate at a reader-free boundary

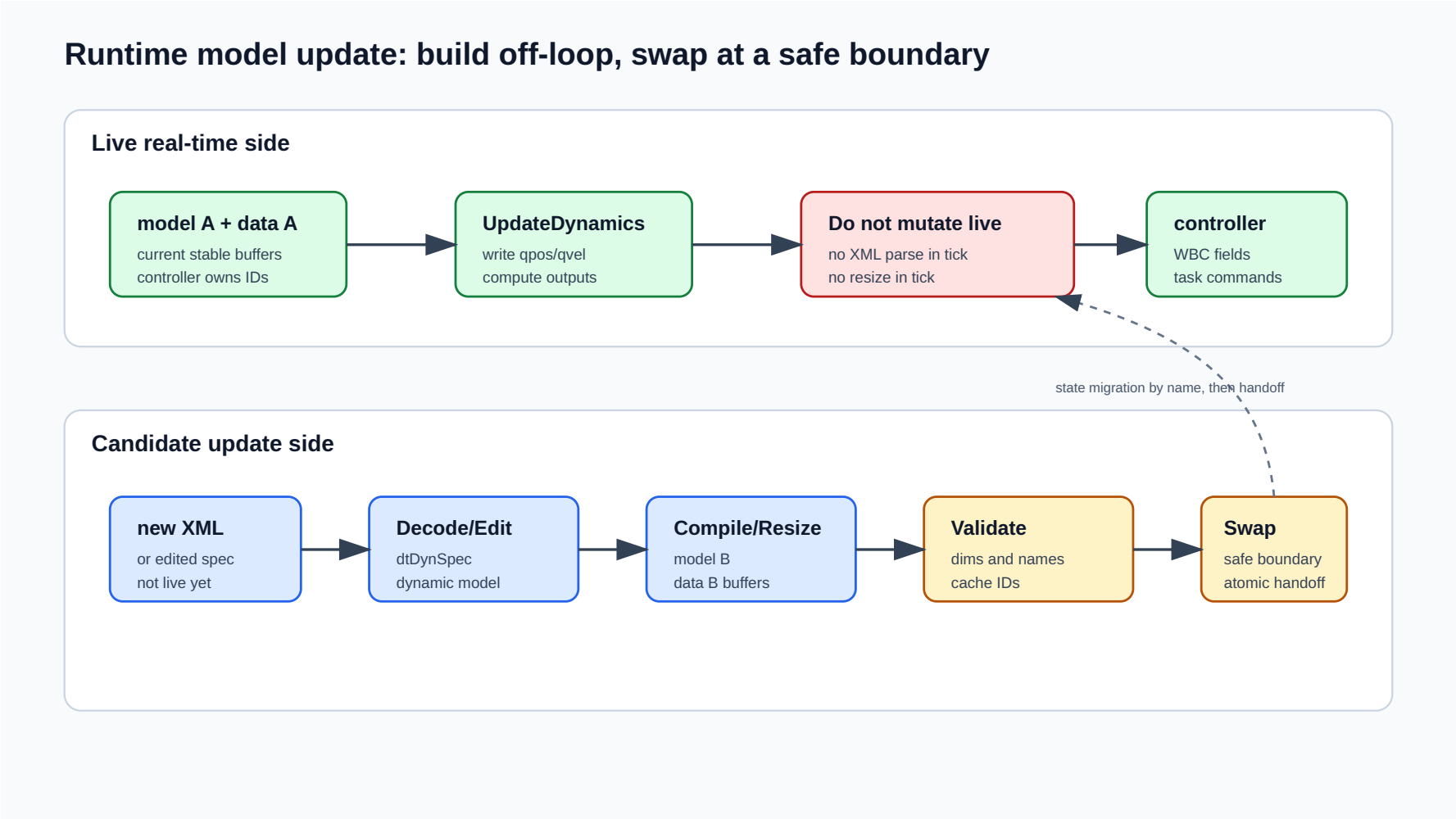
Two safe model-change routes

> Never update a `dtDynModel` while the control loop reads it. Fixed-layout physical changes use validated `UpdateSpec` only at a reader-free control-safe boundary. Structural changes build an independent model/data candidate off the real-time path, then swap it at a safe boundary.

A structural candidate must pass:

- controller dimension checks: `nq`, `nv`
- required body name lookup
- required site name lookup
- `data.IsCompatible(model)` after resize
- rebuilt body/site/Jacobian caches
- pre-sized `dtDynData` and caller-owned outputs on the documented benchmark path
- no XML parsing in the tick; zero counted C++ `new` is limited to the benchmark counter scope, not a whole-process or malloc-family heap guarantee

Runtime model update diagram



Hot-swap container pattern

Proposed pattern — not current LeoQuad production code.

```
struct DynamicsRuntime
{
    dt::Dynamics::dtDynModel model;
    dt::Dynamics::dtDynData data;
    dt::Dynamics::dtDynArray<dt::Dynamics::dtDynIndex> jacBodyIds;
    dt::Dynamics::dtDynArray<dt::Dynamics::dtDynIndex> jacSiteIds;
};

DynamicsRuntime BuildRuntimeFromXml(const std::string &xml)
{
    DynamicsRuntime next;
    const dt::Dynamics::dtDynSpec spec = dt::Dynamics::dtDynSpec::DecodeXml(xml);
    next.model = dt::Dynamics::CompileSpecChecked(spec);
    next.data.Resize(next.model);
    if (!next.data.IsCompatible(next.model))
        throw std::runtime_error("candidate model/data mismatch");
    return next;
}
```

Rule

A safe reload builds an **independent candidate** first.

Then validate the LeoQuad contract before any live swap.

Preserve state by name

When reloading a model with the same named joints:

- map old joint name to new joint ID
- require the same joint type
- copy `qpos` by each joint's `jnt_qposadr`
- copy `qvel` by each joint's `jnt_dofadr`
- skip missing or type-changed joints

> Do **not** assume joint order is unchanged. If a new free joint appears and state is not copied, initialize the quaternion to identity before computing dynamics.

Live swap at a safe boundary

Build and queue the candidate outside the control tick:

```
void RequestModelReload(const std::string &xml)
{
    DynamicsRuntime next = BuildRuntimeFromXml(xml);
    ValidateLeoQuadContract(next.model);
    std::lock_guard<std::mutex> lock(pendingMutex);
    pendingRuntime = std::move(next);
    pendingSwap = true;
}
```

Swap at the control boundary

Migrate state and swap only at the control-thread boundary:

```
std::lock_guard<std::mutex> lock(pendingMutex);
if (pendingSwap) {
    auto &next = pendingRuntime;
    CopyJointStateByName(
        model.dynModel, data.dynData,
        next.model, next.data);
    model.dynModel = std::move(next.model);
    data.dynData = std::move(next.data);
    RebuildLeoQuadCaches(model, data);
    pendingSwap = false;
}
```

For hard real-time code, replace `std::mutex` and heap-owning moves with the process's real-time-safe handoff mechanism. Parse, compile, validate, and allocate outside the tick.

Edit mass with UpdateSpec

`spec` is the editable source; `model` was compiled once and `data` was resized during setup. For a fixed-layout physical update, validate the changed spec before the reader-free control-safe boundary:

```
const dt::Dynamics::dtDynIndex body = spec.BodyId("BODY");
if (body == dt::Dynamics::DTDYN_INVALID_INDEX)
    throw std::runtime_error("required body BODY is missing");

spec.EditBody(body, [](dt::Dynamics::dtDynBodySpec &draft)
              { draft.inertial.mass += 0.25; });

dt::Dynamics::dtDynValidationError error;
if (!dt::Dynamics::ValidateSpec(spec, &error))
    throw std::runtime_error(error.message);

// Synchronize all readers before this control-safe boundary.
if (!dt::Dynamics::UpdateSpec(spec, model))
    throw std::runtime_error("fixed-layout UpdateSpec rejected the spec");
dt::Dynamics::ForwardKinematics(model, data);
```

> A fixed-layout physical edit does not recompile or `Resize`. `UpdateSpec` requires exclusive model access; an incompatible spec returns false without model writes (`false: no model writes`). Its successful write makes cached `dtDynData` results stale, so recompute stale outputs before publication.

Other spec edits: construct bodies and add sites

Create topology programmatically:

```
dt::Dynamics::dtDynSpec spec;
const auto trunk = spec.AddBody("trunk", 0);
spec.AddJoint(
    trunk, "floating_base",
    dt::Dynamics::dtDynJointType::Free);

const auto thigh =
    spec.AddBody("front_left_thigh", trunk);
const auto hip = spec.AddJoint(
    thigh, "front_left_hip",
    dt::Dynamics::dtDynJointType::Hinge);
spec.EditJoint(hip, [](dt::Dynamics::dtDynJointSpec &draft)
               {
                   draft.axis.SetZero();
                   draft.axis(1) = 1.0;
               });
```

Add a site and rebuild caches

```
const auto site =
    spec.AddSite(thigh, "BODY_DEBUG_SITE");
spec.EditSite(site, [](dt::Dynamics::dtDynSiteSpec &draft)
              { draft.pos(2) = 0.2; });

auto model = dt::Dynamics::CompileSpecChecked(spec);
dt::Dynamics::dtDynData data;
data.Resize(model);
// No neutral-state initializer: stage a valid free-joint pose.
data.qpos(3) = 1.0; // identity quaternion w

dt::Dynamics::ForwardKinematics(model, data);

const auto siteId =
    model.SiteId("BODY_DEBUG_SITE");
if (siteId == dt::Dynamics::DTDYN_INVALID_INDEX) {
    throw std::runtime_error("required site missing");
}
dt::Dynamics::ComputeSiteJacobian(
    model, data, siteId, jacp, jacr);
```

Adding a site changes `nsite` but not `nq` or `nv`; it still requires compile and resize because site cache arrays change.

Common mistakes

Mistake	Correct approach
Reparse XML every control tick	Parse/compile once, update `qpos/qvel` each tick
Edit `dtDynModel` arrays directly	Fixed-layout physical value: edit `dtDynSpec`, validate, `UpdateSpec` at an exclusive boundary, then recompute; structural change: `CompileSpecChecked`, resize data
Keep old cached IDs after reload	Rebuild caches from names
Assume `<geom>` or `<mesh>` affects dynamics	Use `<inertial>`, `<joint>`, `<body>`, `<site>`
Treat body `0` as physical	Body `0` is world
Treat no-joint bodies as nonphysical	No-joint bodies can be fixed physical bodies
Compute sites only when the base Jacobian is needed	Pass the base body ID and site IDs to indexed `ComputeJacobians`
Skip `data.Resize(model)` after a structural edit	Resize, then verify `data.IsCompatible(model)`
Swap while dynamics reads model	Build off-loop, swap at a safe boundary

Validation commands

Standalone `dtDynamics`

```
tools/validate_install_export.sh
tools/validate_package.sh
tools/audit_release_readiness.sh
```

External `leoquad-control`

Not shipped by this package:

```
cmake --build build --target simRobot
tools/validate_simrobot_trot.sh --smoke
tools/validate_simrobot_trot.sh --headless-trot
SIMROBOT_PROFILE_DYNAMICS=1 \
  tools/validate_external_dtdynamics_package.sh
```

Model update review

- XML decodes without equality/tendon rejection.
- `nq` and `nv` match controller expectations.
- required body and task site names resolve.
- cached Jacobian body/site ID lists are rebuilt.
- `data.IsCompatible(model)` confirms dimensions only after resize; IDs/address caches are rebuilt separately.
- the pre-sized benchmarked loop performs no XML work and reports zero counted C++ `new` in the documented counter scope.

One-slide team explanation

1. XML is an import/export format, not the runtime model.
2. `DecodeXml` reads the supported MJCF-like subset into `dtDynSpec`, which is editable and dynamic.
3. `CompileSpecChecked` validates external/editable specs and freezes them into `dtDynModel`; raw `CompileSpec` is reserved for separately validated trusted setup data.
4. `dtDynData` is resized once and reused; each tick updates `qpos/qvel` and recomputes dynamics/Jacobians.
5. Fixed-layout physical changes validate off-loop, call `UpdateSpec` with exclusive model access, then recompute stale outputs. Structural changes are built off-loop, checked-compiled, resized, recached, and swapped at a safe boundary.

> **Rule of thumb** — State changes recompute. Fixed-layout physical changes use `UpdateSpec`. Topology/layout/address/name changes use `CompileSpecChecked` and `Resize`.

Thank you

LeoQuad Control & Dynamics · dtDynamics briefing